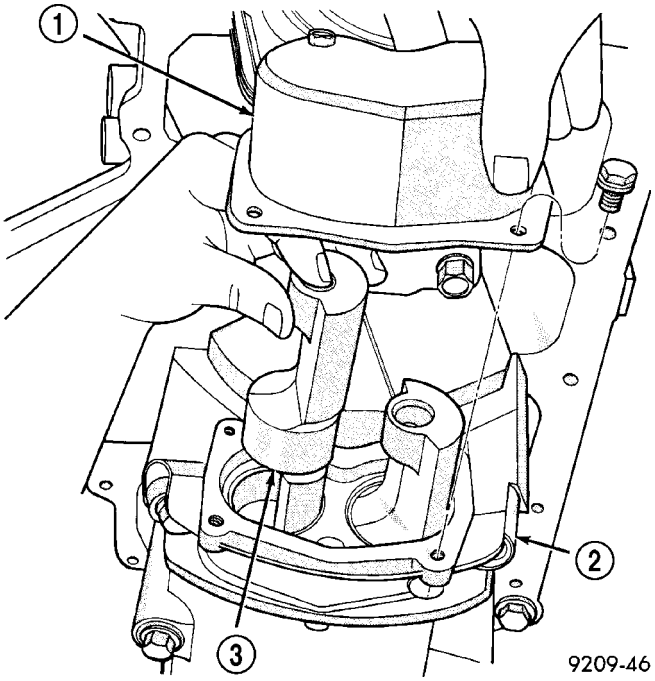


*Fig. 112 Balance Shaft(s)—Removal/Installation*

1 - REAR COVER
2 - CARRIER
3 - BALANCE SHAFT

(8) Remove rear cover and balance shafts (Fig. 112).

(9) Remove four carrier to crankcase attaching bolts to separate carrier from engine bedplate.

## BALANCE SHAFT CARRIER

The following components will remain intact during carrier removal: Gear cover, gears, balance shafts and the rear cover (Fig. 108).

(1) Drain engine oil.

(2) Remove the oil pan and pick-up tube (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).

(3) Remove chain cover, guide and tensioner (Fig. 109).

(4) Remove screw retaining balance shaft drive sprocket (Fig. 110).

(5) Move balance shaft inboard through drive chain sprocket. Sprocket will hang in lower chain loop.

(6) Remove carrier to crankcase attaching bolts to remove carrier.

# INSTALLATION

## BALANCE SHAFT INSTALLATION / TIMING

Balance shaft and carrier assembly installation is the reverse of the removal procedure. **During installation crankshaft-to-balance shaft timing must be established. Refer to Timing procedure in this section.**

(1) With balance shafts installed in carrier (Fig. 108) position carrier on crankcase and install four attaching bolts and tighten to 54 N·m (40 ft. lbs.).

(2) Turn balance shafts until both shaft key ways are up, parallel to vertical centerline of engine. Install short hub drive gear on sprocket driven shaft and long hub gear on gear driven shaft. After installation gear and balance shaft keyways must be up with gear timing marks meshed as shown in (Fig. 113).

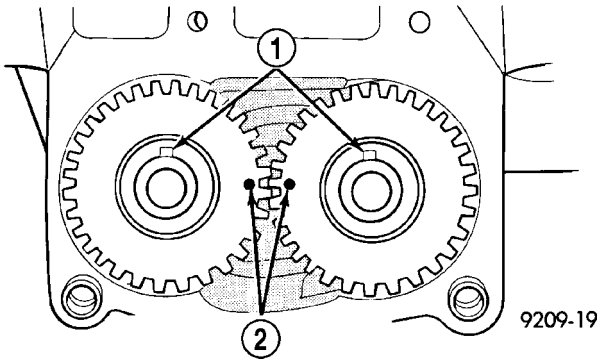


*Fig. 113 Gear Timing*

1 - KEY WAYS UP
2 - GEAR ALIGNMENT DOTS

(3) Install gear cover and tighten double ended stud/washer fastener to 12 N·m (105 in. lbs.).

(4) Align flat on balance shaft drive sprocket to the flat on crankshaft (Fig. 114).

(5) Install balance shaft drive sprocket on crankshaft using Special Tool 6052 (Fig. 115).

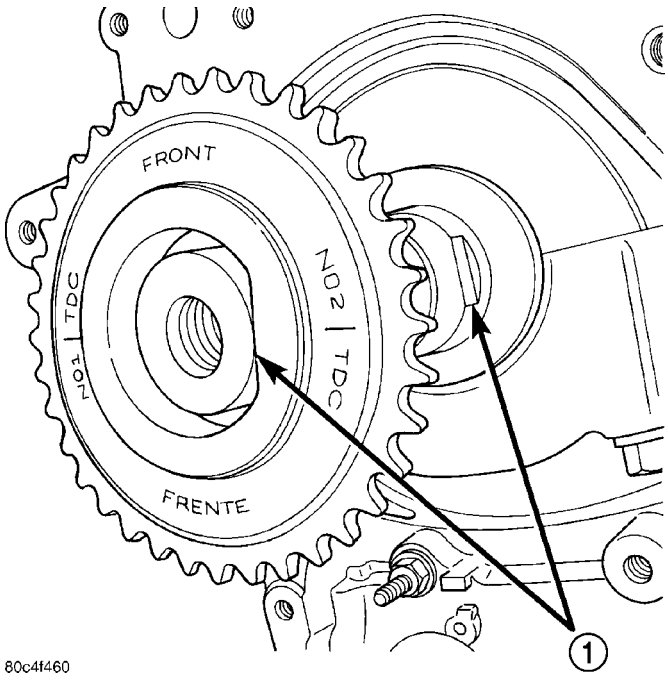


*Fig. 114 Balance Shaft Sprocket Alignment to Crankshaft*

1 - ALIGN FLATS